

SAI ZAYAR HEIN

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Education

King Mongkut's University of Technology Thonburi

Expected Graduation: Jul 2028

Bachelor of Science in Computer Science

Bangkok, Thailand

- Courses: Software Engineering, Operating Systems, Computer Architecture, Intro to Programming, Algorithms and Data Structures, Object-Oriented Concept, Calculus, Linear Algebra

Work Experience

Freelance

Jul 2025 – Present

Full Stack Engineer

- Engineered **secure multi-location medical facility management system** supporting **15+ interconnected** database entities with **attribute-based access control** scalable to unlimited clinic/pharmacy branches
- Delivered **streamlined healthcare inventory** and **financial operations** by enabling **real-time tracking of 9 medicine types**, **automated invoicing**, and multi-method expense categorization via **ExcelJS** bulk imports/exports
- Developed a **production ready Point of Sale user interface**, featuring **full CRUD operations**, **ABAC**, and **PDF invoice generation**

Backend Web Development Mentor

Jan 2026 – Jul 2026

KMUTT CS Program

Bangkok, Thailand

- Mentored **40+ Computer Science students** in **backend web development** by teaching **Node.js** and **Express** fundamentals, guiding them through building and deploying **RESTful APIs** for course projects

Projects

Unishare (Open Source) | Website | *Source Code*

Next.js | NestJS | PostgreSQL | TypeScript

- Reduced **development overhead** and **CI/CD build times** by **40%** by architecting a **Turborepo monorepo** with **pnpm workspaces** across a **Next.js frontend** and **NestJS backend** with shared **TypeScript configs**
- Achieved **sub-200ms real-time messaging latency** by implementing a **scalable Socket.io system** with **Prisma ORM**, supporting **typing indicators**, **message editing/deletion**, and optimized **React Context updates**

Medical Clinic Point of Sale System | Website | *Source Code*

React | Express | PostgreSQL | TypeScript

- Developed a **full-stack healthcare Point-of-Sale** and management system that **enhanced operational efficiency** for **medical clinics and pharmacies** by implementing **attribute-based access control (ABAC)**, multi-unit inventory tracking, and dynamic financial reporting
- Utilized **PostgreSQL** for data storage, **Express** for a **RESTful API backend**, and **React** for the user interface

Slade Chat | *Source code*

Next.js | NestJS | PostgreSQL | TypeScript

- Engineered a **high-concurrency full-stack messaging platform** with **99.9% message delivery reliability** by implementing a **NestJS** and **Socket.IO** architecture with **Better Auth** integration
- Reduced **primary database write-load** by **50%** by architecting a **Redis-based presence system** to track real-time **online/offline status** and **last-seen timestamps** for active users

Leadership

SIT Hackathon

Jun 2025

- Led a three-member team to complete a **Paradox themed, AI-integrated project** within **28 hours**, designing the system architecture, database, and flow, selecting **Shadcn** as the frontend UI library, and integrating **Gemini** and **Replicate** models for prompt and image generation

Technical Skills

Languages: Java, SQL, HTML5, CSS, JavaScript, TypeScript

Frontend: React, Next.js, Tailwind CSS, Shadcn UI

Backend: Node.js, NestJS, WebSockets (Socket.io), Express.js, Hono, Cloudinary, BullMQ

Databases/ORMs: PostgreSQL, MySQL, SQLite, MongoDB, Redis, Prisma

Infrastructure: AWS, DigitalOcean, Docker, Nginx, Coolify

Developer Tools: Better Auth, Git, Postman